



Key idea (important)

Firebase Authentication already handles:

- Secure password hashing (server-side)
- Salt (automatic per user)
- Pepper (internal system secret, not exposed)
- Slow hashing / key stretching
- Brute-force protection
- Account lockout / rate limiting
- Secure storage (you never see raw passwords or hashes)

👉 So Firebase Auth already behaves like an **industrial-grade password system stronger than most custom implementations**



Comparison for FYP debate

Method	Security level	Real-world usage	Notes
SHA-256 only	✗ Weak	Not used	Too fast
SHA-256 + salt	⚠ Weak-medium	Rare	Still brute-forceable
PBKDF2	⚠ Good	Legacy systems	Acceptable
Argon2id + salt (+ pepper)	★ Very strong	Modern custom systems	Best if self-built
Firebase Authentication	★★ Enterprise-grade	Google-managed systems	Includes multiple layers beyond hashing



Important FYP argument (this is your “winning point”)

You can say:

Firestore Authentication is not just a hashing algorithm — it is a **full authentication system**

It includes:

1. Password hashing (server-side)

You don't control it, but it is:

- salted
- slow hashed

- protected by Google infrastructure

2. Attack protection

- rate limiting
- bot detection
- IP monitoring
- credential stuffing protection

3. Secure architecture advantage

Custom Argon2 system	Firebase Auth
You must secure backend	Google secures it
Risk of implementation mistakes	Minimal
You manage pepper, storage, hashing	Fully abstracted
Security depends on developer	Security depends on Google infrastructure

So in your presentation:

If examiner asks:

“Why not Argon2id instead of Firebase Auth?”

You can answer:

- ✓ Firebase Auth already implements secure password hashing internally
- ✓ It includes salt, key stretching, and server-side protections
- ✓ It reduces human implementation error (which is a major security risk)
- ✓ It adds enterprise-level security features beyond hashing (rate limiting, abuse detection)
- ✓ Therefore, Firebase Auth is not just comparable — it is a **complete authentication framework**

Important clarification

You are NOT comparing:

- ✗ Argon2id vs Firebase hashing algorithm

You are comparing:

- ✓ Custom authentication system vs Managed authentication service

That is why Firebase usually “wins” in real-world deployment.

Final conclusion

- You do **NOT** need SHA-256 + Argon2 + salt + pepper together
- If using Firebase Auth → **do nothing extra** for password hashing
- If building your own system → use **Argon2id + salt** (+ optional pepper)